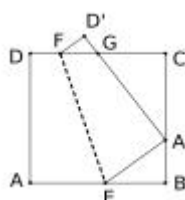


INMO(32nd) – 2017

Time – 4 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
 - All Questions carry equal marks. Maximum marks: 102.
 - Answer to each question should start on a new page. Clearly indicate the question number.
1. In the given figure, ABCD is a square sheet of paper. It is folded along EF such that A goes to a point A' different from B and C, on the side BC and D goes to D' . The line $A'D'$ cuts CD in G. Show that the inradius of the triangle GCA' is the sum of inradii of the triangle $GD'F$ and $A'BE$.



2. Suppose $n \geq 0$ is an integer and all the roots of $x^3 + \sim x + 4 - (2 \times 2016^n) = 0$ are integers. Find all possible values of $\sim$.
3. Find the number of triples (x, a, b) where x is a real number and a, b belongs to the set $\{1, 2, \dots, 9\}$ such that $x^2 - a\{x\} + b = 0$, where $\{x\}$ denotes the fractional part of the real number x . For example $\{1.1\} = \{-0.9\} = 0.1$.
4. Let ABCDE be a convex pentagon in which $\angle A = \angle B = \angle C = \angle D = 120^\circ$ and side lengths are five consecutive integers in some order. Find all possible values of $AB + BC + CD$.
5. Let ABC be a triangle with $\angle A = 90^\circ$ and $AB < AC$. Let AD be the altitude from A onto BC. Let P, Q and I denote respectively the incentres of triangles ABD, ACD and ABC. Prove that AI is perpendicular to PQ and $AI = PQ$.
6. Let $n \geq 1$ be an integer and consider the sum:

$$x = \sum_{k \geq 0} \binom{n}{2k} 2^{n-2k} 3^k = \binom{n}{0} 2^n + \binom{n}{2} 2^{n-2} 3 + \binom{n}{4} 2^{n-4} 3^2 + \dots$$

Show that $2x-1, 2x, 2x+1$ forms the sides of a triangle whose area and inradius are also integers.

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